Harbinger Route Calculator: Engineering a High-Performance Telemetry Engine

Data Engineering Team

2026-03-24

Table of contents

1 Executive Summary **1**

2 1. Performance Optimization: Overcoming the Python Bottleneck **2**
2.1 The Original Bottleneck: `iterrows()` and Scalar Math 2
2.2 The Solution: Vectorization and `itertuples()` 2
2.2.1 Vectorized Math Implementation 2

3 2. Core Route Engine Logic **3**
3.1 The Split Triggers 3

4 3. Fault Tolerance: The Sliding Window & Reconciliation **3**
4.1 Deterministic UUIDs 4
4.2 Window Reconciliation Logic 4
4.3 PostgreSQL UPSERT Integrity 5

5 4. Visualizing Fleet Operations **5**
5.1 Raw Route Telemetry (Database View) 5
5.2 Fleet Route Dashboard (Grafana) 6

1 Executive Summary

The Harbinger Route Calculator processes millions of rows of synchronized CAN bus telemetry to generate deterministic, consolidated driving routes for the fleet.

In its latest iteration, the calculation engine underwent a massive architectural refactor to eliminate Python-level memory allocation overhead and database query bottlenecks. By

leveraging **C-level Pandas vectorization**, lightweight iteration, and bulk PostgreSQL upserts, processing time for a standard 180-truck batch dropped from ~**48 seconds to ~0.7 seconds**—a **64× performance improvement**.

---

2 1. Performance Optimization: Overcoming the Python Bottleneck

2.1 The Original Bottleneck: `iterrows()` and Scalar Math

The initial engine processed telemetry sequentially using Pandas `iterrows()`. Because Python is dynamically typed, iterating row-by-row forces the interpreter to perform type checks, allocate memory for a full Pandas Series object, and dispatch mathematical operations sequentially for every single row. Across 500,000+ rows of telemetry, this CPU thrashing caused severe latency.

2.2 The Solution: Vectorization and `itertuples()`

To achieve sub-second processing, the engine was rewritten to utilize **vectorization**.

Instead of computing time gaps and distance deltas inside a Python loop, these operations are now applied to entire columns simultaneously. This pushes the heavy mathematical lifting down to highly optimized C/C++ execution paths at the hardware level.

Once the math is calculated in bulk, the engine iterates over the data using `itertuples()`, which yields lightweight, read-only NamedTuples with near-zero overhead.

2.2.1 Vectorized Math Implementation

```
# --- Vectorization (The Math) ---
# Executed in bulk via C-level paths before iteration begins
df['odo_mi'] = df['odometer'] * METERS_TO_MILES
df['time_gap'] = df['time'].diff().dt.total_seconds().fillna(0.0)

if 'prnd_state' not in df.columns:
    df['prnd_state'] = 0  # Default to 0 if column is missing

# --- Lightweight Iteration ---
for row in df.itertuples():
```

```
    # Dot-notation access avoids dictionary-style lookups
    t = row.time
    odo_mi = row.odo_mi
    soc = row.soc
    charge_st = row.charge_state
    prnd_st = row.prnd_state
    time_gap = row.time_gap
    # ... logic continues ...
```

---

3 2. Core Route Engine Logic

The engine must intelligently parse continuous streams of telemetry to determine when a route begins and, more importantly, *why* it ends.

A route actively starts when the vehicle registers its first odometer increase (`> 0.01 miles`) while the vehicle is not in Park (`prnd_state != 1`).

3.1 The Split Triggers

The engine determines the end of a route based on three primary operational states:

1. **Charge:** The route naturally ends because the truck was plugged in. To prevent signal noise from triggering false stops, this state requires `charge_state == 8` for $k \geq 15$ consecutive records (the noise filter).
2. **Idle:** The route ends because the truck parked and stopped moving. Triggered when a time gap of ≥ 2.5 hours occurs between valid telemetry rows.
3. **Open:** The truck was still actively driving when the 48-hour data query window cut off. This route is saved to the database but flagged as unfinished, awaiting the next batch run.

---

4 3. Fault Tolerance: The Sliding Window & Reconciliation

Because the engine processes data in 48-hour batches, a truck may be actively driving precisely when the time window cuts off resulting in an **"Open"** route. To prevent duplicating this trip when the next batch runs, the engine employs a highly fault-tolerant overlap reconciliation strategy.

4.1 Deterministic UUIDs

Primary keys for routes are generated dynamically using a namespace hash of the `deviceid` and the exact `start_time`.

4.2 Window Reconciliation Logic

Before finalizing the calculated routes, the engine queries the PostgreSQL database for any historical routes that overlap with the current processing window.

If a newly calculated route overlaps with an existing database route (especially an "Open" one), the engine **inherits** the original start time and start odometer from the database.

```
# --- WINDOW RECONCILIATION ---
final_routes = []
for r in raw_routes:
    calc_start = pd.to_datetime(r['start_time']).tz_localize(None)
    calc_end = pd.to_datetime(r['end_time']).tz_localize(None)

    matched_db_route = None

    # Check against overlapping routes pulled from the database
    if existing_routes:
        for db_r in existing_routes:
            db_start = pd.to_datetime(db_r['start_time']).tz_localize(None)
            db_end = pd.to_datetime(db_r['end_time']).tz_localize(None) if pd.notnull(db_r.ge

            if calc_start <= db_end and calc_end >= db_start:
                matched_db_route = db_r
                break

    if matched_db_route:
        # INHERIT FROM DATABASE to prevent duplicates
        r['start_time'] = pd.to_datetime(matched_db_route['start_time'])
        r['start_odo'] = matched_db_route['start_odo']
        r['soc_start'] = matched_db_route.get('soc_start', r['soc_start'])

    r['miles_driven'] = r['end_odo'] - r['start_odo']
    r['id'] = generate_uuid(r['deviceid'], r['start_time'])

    if r['split_reason'] == 'Open' or r['miles_driven'] >= min_miles:
        final_routes.append(r)
```

4.3 PostgreSQL UPSERT Integrity

Because the UUID is strictly deterministic, writing the reconciled route to the database triggers an `ON CONFLICT` clause. Instead of failing or duplicating, PostgreSQL seamlessly updates the `end_time`, `end_odo`, and `split_reason` of the original route.

```
INSERT INTO driving_route (
    id, deviceid, start_time, end_time,
    start_odo, end_odo, miles_driven,
    soc_start, soc_end, split_reason
)
VALUES (
    :id, :deviceid, :start_time, :end_time,
    :start_odo, :end_odo, :miles_driven,
    :soc_start, :soc_end, :split_reason
)
ON CONFLICT (id)
DO UPDATE SET
    end_time = EXCLUDED.end_time,
    start_odo = EXCLUDED.start_odo,
    end_odo = EXCLUDED.end_odo,
    miles_driven = EXCLUDED.miles_driven,
    soc_end = EXCLUDED.soc_end,
    split_reason = EXCLUDED.split_reason,
    recordupdatedat = NOW();
```

---

5 4. Visualizing Fleet Operations

The raw data calculated by the engine provides the foundation for fleet analytics, range estimation, and operational monitoring.

5.1 Raw Route Telemetry (Database View)

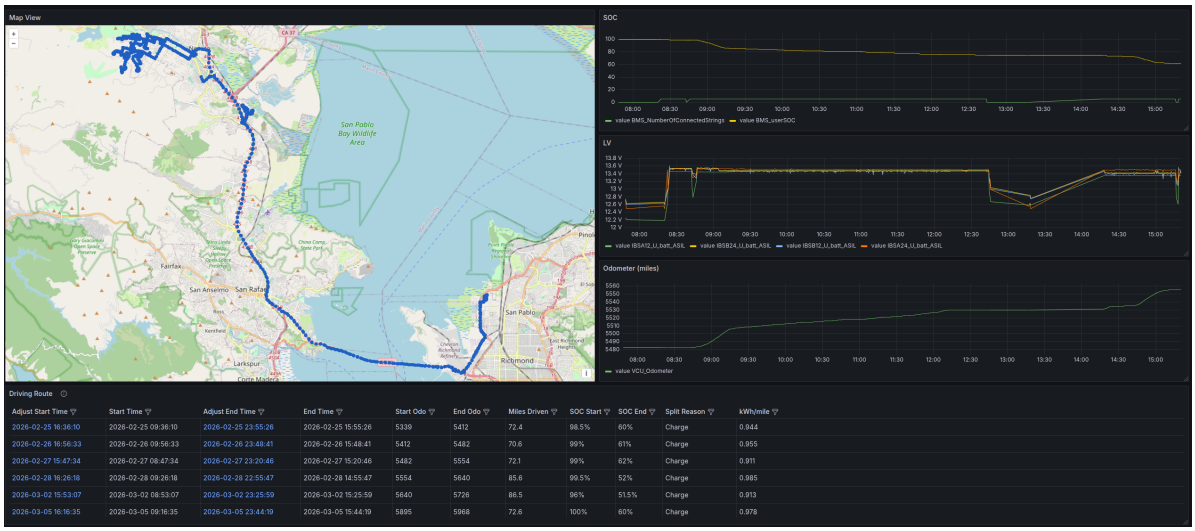
Below is a sample of the consolidated data stored in the `driving_route` table, representing the fully reconciled trips.

	id	deviceid	start_time	end_time	start_odo	end_odo	miles_driven	soc_start	soc_end	split_reason	recordupdatedat	recordcreatedat
1	b9424e36-897e-59d8-85d6-741d5e6f87e9	004cf7c2-1fea-43f3-8706-226172b78823	2026-02-26 16:45:31.000	2026-02-26 18:26:52.000	4.22	6.85	2.64	52.5	49.5	Idle	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
2	04cd2e9a-157f-5de1-914c-5f721e46744e	03d69f35-f4cc-4531-9152-60b2ff437078	2026-02-26 17:06:29.000	2026-02-26 17:12:09.000	5.37	7.41	2.03	61	59.5	Idle	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
3	095cb4b6-c44a-557f-b24d-9b9ce97659a2	074be5c3-5c24-4aff-8b02-7f355bc2dd1d	2026-02-27 12:35:19.000	2026-02-27 13:55:09.000	10.27	13.73	3.46	76.5	74.5	Idle	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
4	b84f908c-db4d-533e-a191-eb32d2f5ec7b	0d83333a-0a6f-4a78-9f56-032a884ed350	2026-02-25 15:42:38.000	2026-02-25 20:23:01.000	57.51	79.4	21.89	79	59.5	Idle	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
5	40e79b6f-8435-5e5f-be49-fe7b1d61cf11	1476a5d4-1070-4e71-84dc-fe2fa0db3e6a	2026-02-26 15:48:56.000	2026-02-26 16:43:07.000	833.58	840.45	6.88	88	82	Charge	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
6	2ca8ef7a-5c38-5e6c-95ca-ba32605504e4d	1476a5d4-1070-4e71-84dc-fe2fa0db3e6a	2026-02-26 18:28:12.000	2026-02-26 20:00:04.000	840.49	857.83	17.35	82	74.5	Idle	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
7	359b833d-15d7-585b-8063-bd40f09662b0	227eaa68-6aa2-4991-a2c8-e495c54ed9f2	2026-02-25 16:16:30.000	2026-02-25 20:12:21.000	188.8	194.24	5.44	57	54	Idle	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
8	5ef520a7-0c46-518a-9296-476be6824604	309e63ef-5fe7-4c79-acd6-020f71b6e03d	2026-02-26 14:08:22.000	2026-02-26 14:42:42.000	5.58	30.05	24.46	96.5	77.5	Idle	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
9	d74aad5d-fbe9-54f7-913f-030f31648e04	309e63ef-5fe7-4c79-acd6-020f71b6e03d	2026-02-26 19:43:24.000	2026-02-26 19:50:44.000	30.08	32.34	2.26	77.5	76	Idle	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
10	e4363b7f-8ca5-5ce9-8125-4ae509c47d8b	35aac802-cb22-4074-8fe0-ddb2f76425b2	2026-02-25 16:43:28.000	2026-02-25 18:38:18.000	0.01	1.78	1.78	25	24	Open	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
11	54ff0c2-2575-5f1e-aeb6-b308aebde31b	3a5a4e6d-0c8f-4334-97ad-fdaa9d5c6377	2026-02-25 16:23:49.000	2026-02-25 17:07:30.000	2,034.37	2,046.29	11.92	53	46	Charge	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
12	c679f888-e461-5511-912f-a6e29ed371ea	3a5a4e6d-0c8f-4334-97ad-fdaa9d5c6377	2026-02-25 18:03:52.000	2026-02-25 18:36:22.000	2,046.74	2,055.7	8.97	60.5	56	Charge	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
13	ba45bd35-8fb2-5075-ac42-2562920ca56a	3a5a4e6d-0c8f-4334-97ad-fdaa9d5c6377	2026-02-25 19:08:54.000	2026-02-25 19:52:25.000	2,055.71	2,070.7	14.98	71	63.5	Charge	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
14	d1ffbf4e-a165-5878-b57f-00f8a93e74d7	3a5a4e6d-0c8f-4334-97ad-fdaa9d5c6377	2026-02-25 20:45:02.000	2026-02-25 23:57:31.000	2,070.74	2,132.53	61.79	84.5	49	Charge	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
15	73083411-d516-56c1-9821-92333f20d375	3a5a4e6d-0c8f-4334-97ad-fdaa9d5c6377	2026-02-26 14:14:40.000	2026-02-26 16:58:53.000	2,133.07	2,149.62	16.56	79	69	Idle	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
16	77545144-ac6b-54af-8974-18deb35f84e6	3cb437f1-f23f-4b39-bcfe-344cfab332fe	2026-02-25 15:32:43.000	2026-02-25 21:08:19.000	82.85	177.52	94.67	95	40	Charge	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
17	c1c024d6-bb5d-5c0d-abce-5543b4d928db	3cb437f1-f23f-4b39-bcfe-344cfab332fe	2026-02-25 22:35:31.000	2026-02-26 00:40:32.000	177.52	273.96	96.44	97.5	31	Idle	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
18	cd61a198-9c80-5004-9a3e-83547e50a7b1	3cb437f1-f23f-4b39-bcfe-344cfab332fe	2026-03-06 17:44:35.000	2026-03-07 01:02:05.000	424.03	504.86	80.83	99.5	49	Charge	2026-03-08 00:15:03.927	2026-03-06 18:15:04.933
19	6d0ca055-0bb9-55de-9894-55856187e8b6	3e16ee6f-b5d0-434f-8644-3aba6b9daf9c	2026-02-25 19:26:22.000	2026-02-25 19:38:52.000	0.02	1.87	1.85	24.5	24	Open	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
20	82492a29-fc72-562c-8ef6-cc8649bca273	1fb3ab2e-df65-4ff0-8aed-d28381d0161d	2026-03-10 13:53:53.000	2026-03-10 14:12:13.000	2.31	4.32	2.01	99.5	98.5	Idle	2026-03-11 13:15:04.238	2026-03-10 14:15:05.722
21	7ef513fc-e807-590f-bab4-37cb9cbd8b56	4a9ec099-8390-43b4-83fa-4eed486db6c9	2026-02-25 23:17:32.000	2026-02-25 23:38:23.000	3,662.02	3,667.33	5.31	83	80.5	Idle	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
22	953de431-383c-5531-bd6f-41ff6a8c3c59	50feec7e-92da-4028-86d2-62f448dcb65a	2026-02-25 17:36:10.000	2026-02-25 23:55:26.000	5,339.2	5,411.57	72.37	98.5	60	Charge	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
23	4de39400-cf8c-54d6-a45c-dadcb5a04743	50feec7e-92da-4028-86d2-62f448dcb65a	2026-02-26 17:56:33.000	2026-02-26 23:48:41.000	5,411.57	5,482.17	70.6	99	61	Charge	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
24	e565934d-0a5e-532f-89e8-f3a6be7c2bbf	3a5a4e6d-0c8f-4334-97ad-fdaa9d5c6377	2026-03-06 14:16:40.000	2026-03-06 18:02:44.000	2,291.88	2,300.98	9.1	31.5	27	Idle	2026-03-07 17:15:03.645	2026-03-06 18:15:04.933
25	d1c9667d-5ce1-5101-bf85-f831d1fdd6dc	5e265628-f85c-423f-bb93-16bf71f929be	2026-02-27 00:19:11.000	2026-02-27 00:38:42.000	175.46	177.46	2	49.5	48	Charge	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
26	4c644101-8c42-5cf3-9968-630b15d72bf8	69272013-f775-4c20-bd86-7886d32d40f3	2026-02-26 20:46:05.000	2026-02-26 21:37:36.000	7.2	37.41	30.22	45	37.5	Idle	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
27	c505d08f-0edb-5b78-8f4a-b28a19bd9172	7843ec48-9c41-4519-822a-17dcf75b86c0	2026-02-26 18:24:02.000	2026-02-26 19:26:40.000	4	6.42	2.41	79	74.5	Idle	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
28	05fe55f0-f53d-53a7-8bc9-0b65288d2a8f	78786e06-0637-4e6a-8a9a-703c2b2ffdfa	2026-02-25 14:41:42.000	2026-02-25 22:12:49.000	2,217.77	2,313.89	96.12	88	36.5	Idle	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
29	feb63da8-d7e4-56b8-b7e0-56a0d8dff023	78786e06-0637-4e6a-8a9a-703c2b2ffdfa	2026-02-26 14:20:44.000	2026-02-26 21:42:46.000	2,313.93	2,354.57	40.65	81	60	Charge	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
30	020ed47d-a39e-5759-b0a8-0563966c2b83	89a0c51b-653e-4df1-945c-99bc8219aec2	2026-02-26 18:51:34.000	2026-02-26 19:21:57.000	3.97	6.29	2.32	72	69	Idle	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
31	2d021fa3-05e6-5f5e-9304-1d904eab3e01	9073a1b2-10de-4b0b-abe7-ca476cac53fc	2026-02-27 15:48:28.000	2026-02-27 18:20:49.000	4.06	6.57	2.51	99	93.5	Idle	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
32	5d2fb224-62d9-520b-8300-d1220b5886fa	92144019-f258-4c7f-8952-ac46ff67c2fa	2026-02-25 21:52:55.000	2026-02-26 04:09:49.000	3,020.08	3,067.91	47.83	63	35.5	Idle	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
33	0a3e5ebd-0245-5f49-8c26-50f3ae4da4c3	92144019-f258-4c7f-8952-ac46ff67c2fa	2026-02-26 15:22:39.000	2026-02-26 19:40:59.000	3,067.91	3,101.5	33.59	35.5	17	Charge	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
34	623861f9-e71a-5672-ba7a-1bca9bf5a426	92144019-f258-4c7f-8952-ac46ff67c2fa	2026-02-26 21:20:21.000	2026-02-26 23:49:08.000	3,101.51	3,106.79	5.28	26	22	Charge	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
35	e0e18b95-9250-552d-854c-645c5ab97833	a0851cd0-f151-4623-a50b-e4797c3c0b1c	2026-02-25 19:58:55.000	2026-02-25 20:18:55.000	0.02	1.83	1.81	22.5	21.5	Open	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
36	e9b5fcd8-e07a-5414-abb6-5b55ef11fe7b	227eaa68-6aa2-4991-a2c8-e495c54ed9f2	2026-03-06 17:37:22.000	2026-03-07 02:57:42.000	401.63	466.79	65.16	87.5	43.5	Idle	2026-03-08 03:15:03.615	2026-03-06 18:15:04.933
37	567dca90-634d-5a8b-80f0-0c6eab3d88e7	aad793d7-4419-484e-a0a8-3cffd35341c5	2026-02-26 16:43:24.000	2026-02-26 23:18:44.000	2,699.86	2,747.55	47.69	64	25	Charge	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
38	6d327265-5f36-5b79-a36d-ead56aee9453	aad793d7-4419-484e-a0a8-3cffd35341c5	2026-02-27 16:48:16.000	2026-02-27 17:02:18.000	2,747.64	2,749.82	2.18	34	34	Charge	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
39	9a57488f-c1b1-5759-9a2a-debb6e658bd9	50feec7e-92da-4028-86d2-62f448dcb65a	2026-03-06 17:10:08.000	2026-03-07 00:26:25.000	5,968.08	6,046.55	78.47	99.5	58	Charge	2026-03-07 23:15:03.416	2026-03-06 18:15:04.933
40	ba1b76e2-edf3-5d9e-9a2e-e16964e9f92f	4d340271-c087-4133-9d74-23742ab7a445	2026-03-05 22:44:41.000	2026-03-06 01:03:35.000	1,015.94	1,056.91	40.97	61.5	42.5	Idle	2026-03-07 00:15:04.364	2026-03-05 23:15:03.699
41	60392565-21d7-5c60-be3f-d97b7f91640e	4a9ec099-8390-43b4-83fa-4eed486db6c9	2026-03-10 15:30:18.000	2026-03-10 17:22:41.000	3,703.12	3,745.2	42.08	96	80	Charge	2026-03-11 15:15:04.718	2026-03-10 16:15:06.302
42	36e0ccbc-8b7d-572d-bb1b-13f3f90e59a5	92144019-f258-4c7f-8952-ac46ff67c2fa	2026-03-14 03:55:40.000	2026-03-14 04:06:31.000	3,533.42	3,536.84	3.42	52	49.5	Idle	2026-03-15 03:15:03.367	2026-03-14 04:15:04.308
43	777a1c41-ec85-5b66-8c8f-021c94b54201	227eaa68-6aa2-4991-a2c8-e495c54ed9f2	2026-03-10 04:18:15.000	2026-03-10 04:26:35.000	475.38	478.58	3.2	96	92.5	Idle	2026-03-11 04:15:05.220	2026-03-10 05:15:06.262
44	94337042-320d-5523-b91a-f09830b14122	072ad260-6cd3-4793-89af-74ce6afba38c	2026-03-12 18:43:48.000	2026-03-12 19:38:45.000	296.69	305.06	8.37	97.5	92	Idle	2026-03-14 01:40:59.492	2026-03-12 19:15:05.498
45	1e40c1e9-101c-55fc-b74b-c61fc993254f	b389feb5-b10f-4b42-87dd-62bd756cd2cc	2026-02-25 18:48:40.000	2026-02-25 19:08:31.000	0.04	1.92	1.88	24.5	23.5	Open	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
46	b0fbab3d-af45-5b9e-a890-08959d3587a7	b5f98353-c811-484b-ab7f-3b51af4a4713	2026-02-25 17:31:28.000	2026-02-26 00:02:01.000	5,968.17	6,038.36	70.18	99	62.5	Charge	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
47	a46f7616-2ee2-58d8-ae27-5b09e0fdd7ce	b5f98353-c811-484b-ab7f-3b51af4a4713	2026-02-26 17:57:13.000	2026-02-26 23:11:44.000	6,038.36	6,111.73	73.37	98.5	59.5	Charge	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
48	d3b0c3ed-c669-5448-a675-bbd7b680a236	c250af05-b361-4b95-94be-815fc8a01d46	2026-02-26 17:24:19.000	2026-02-27 00:08:13.000	4,951.78	5,010.09	58.3	63	25.5	Charge	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
49	e29ac928-c0f5-5391-b0a3-4ce7a9c5653d	c250af05-b361-4b95-94be-815fc8a01d46	2026-02-27 01:42:15.000	2026-02-27 01:58:36.000	5,010.09	5,013.6	3.51	99	93	Idle	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
50	c85f00d9-719f-52b2-855e-195d95c6d23	cb750513-5ffa-4d02-986c-0ad72ddc54e0	2026-02-25 00:48:37.000	2026-02-25 15:30:24.000	2,204.07	2,209.58	5.51	97.5	91.5	Idle	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
51	0140a021-0ebb-5796-8043-d41415ab4e51	cb750513-5ffa-4d02-986c-0ad72ddc54e0	2026-02-25 15:30:24.000	2026-02-26 17:20:35.000	2,209.62	2,229.94	20.31	91.5	77.5	Idle	2026-02-28 00:28:32.692	2026-02-28 00:28:32.692
52	95f56e57-39d6-5258-a988-702e706d157f	1d51be9f-a42a-4f19-b98c-0088514b4ee8	2026-03-13 12:46:46.000	2026-03-13 12:54:47.000	6.53	8.4	1.87	52	50.5	Idle	2026-03-14 12:15:04.181	2026-03-13 13:15:04.230
53	3f263b17-ba29-5166-b2e2-604816a673fe	5e265628-f85c-423f-bb93-16bf71f929be	2026-03-06 19:56:57.000	2026-03-07 01:21:50.000	182.37	184.86	2.49	43.5	42	Idle	2026-03-07 21:15:03.477	2026-03-06 23:15:05.047

Figure 1: Raw Driving Route Database Table

5.2 Fleet Route Dashboard (Grafana)

By parsing the deterministic route records, we can visualize active vehicle states, historic trips, and charging behaviors across the entire fleet in near-real-time.



Adjust Start Time	Start Time	Adjust End Time	End Time	Start Odo	End Odo	Miles Driven	SOC Start	SOC End	Split Reason	kWh/mile
2026-02-25 16:36:10	2026-02-25 09:36:10	2026-02-25 23:55:28	2026-02-25 15:55:28	5339	5412	72.4	98.5%	60%	Charge	0.944
2026-02-26 16:56:33	2026-02-26 09:56:33	2026-02-26 23:48:41	2026-02-26 15:48:41	5412	5482	70.6	99%	61%	Charge	0.955
2026-02-27 15:47:34	2026-02-27 08:47:34	2026-02-27 23:20:46	2026-02-27 15:20:46	5482	5554	72.1	99%	62%	Charge	0.911
2026-02-28 16:26:18	2026-02-28 09:26:18	2026-02-28 22:55:47	2026-02-28 14:55:47	5554	5640	85.6	99.5%	52%	Charge	0.985
2026-03-02 15:53:07	2026-03-02 08:53:07	2026-03-02 23:25:59	2026-03-02 15:25:59	5640	5726	86.5	96%	51.5%	Charge	0.913
2026-03-05 16:16:35	2026-03-05 09:16:35	2026-03-05 23:44:19	2026-03-05 15:44:19	5895	5968	72.6	100%	60%	Charge	0.978

Figure 2: Fleet Route Visualization in Grafana